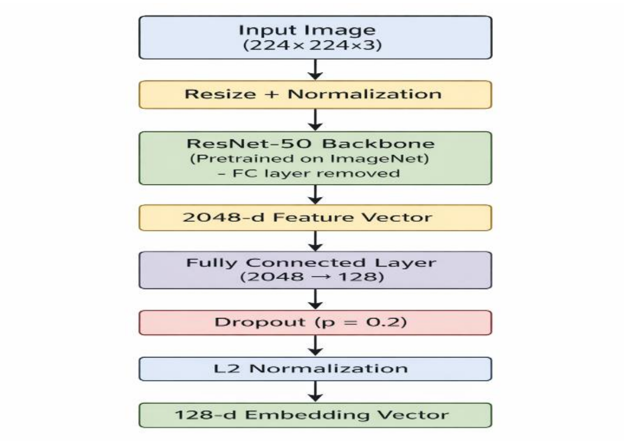
MSDS25008 **Neeka Javed**

Deep Metric Learning for Image Retrieval

Model Architecture: Diagram and Design Decisions

In this research, a pre-tained ResNet-50 backbone serves as the foundation for a deep metric learning network. The network maps each input image into a 128-dimensional embedding space, where images from different classes are far apart and images from the same class are near together



<u>Summary of Implementation</u>

A pretrained ResNet-50 from Torchvision is used to implement the model. A bespoke embedding layer takes the place of the original classification head. In order to make similarity comparisons more reliable and significant, the output embedding is normalized.

Design Decisions

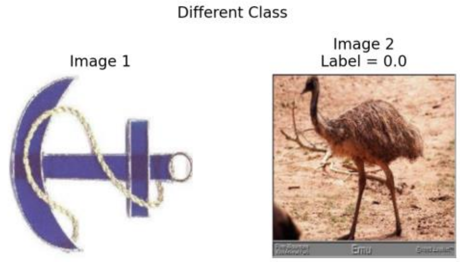
1. ResNet-50 as the foundation
Because ResNet-50 is a powerful and popular convolutional neural network for visual feature extraction, it was chosen. Instead of learning everything from scratch, the model can begin with rich visual information by using pretrained ImageNet weights.
2. **Eliminating the layer of classification**
ResNet-50's initial fully linked layer was created for ImageNet categorization. That layer is eliminated and replaced with an embedding projection layer because the goal of this research is picture retrieval rather than classification.
3. **An embedding layer with 128 dimensions**
the 2048-dimensional ResNet output is mapped to a 128-dimensional embedding using a linear layer. Compact, effective, and appropriate for distance-based retrieval is this lower-dimensional space.
4. **Regularization of dropouts**
to lessen overfitting and enhance generalization, a dropout layer with probability 0.2 is applied after the linear layer.

5. **Normalization of L2**
Unit length is used to normalize the final embedding. Because retrieval depends on the similarity or distance between embeddings, this is crucial. Normalization enhances metric-learning behavior and increases consistency in comparisons.
6. **Layers of frozen backbone**
during training, only the embedding layer is trainable; all other ResNet-50 backbone parameters are fixed. This keeps the final representation tailored to the retrieval problem while cutting down on training time and computing expense.
Why this architecture is appropriate
this architecture is well suited for image retrieval because it does not predict class labels directly. Rather, it learns a feature space that maintains semantic similarity. It is therefore useful for:

- nearest-neighbor search
- Recall@K assessment
- Visualization of t-SNE
- retrieval of images that are visually comparable

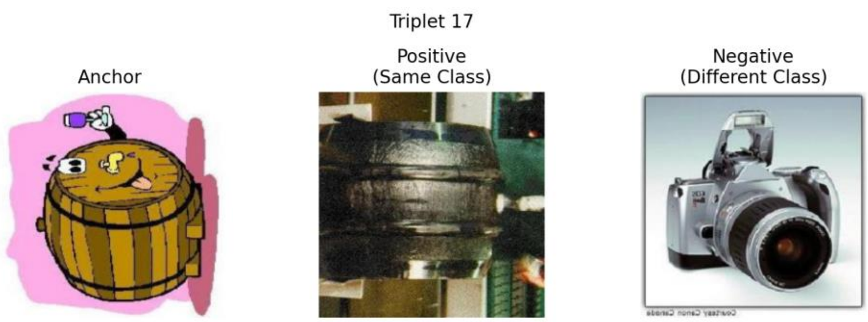
Method for Sampling Datasets: Creating Pairs and Triplets
In order to serve various metric learning purposes, the dataset is processed and sampled in an organized way in this project. After splitting the dataset into training, validation, and testing sets, certain sampling techniques are used for triplet and contrastive learning.
Splitting Datasets
Three subsets are created from the dataset:
70% of the training set
15% is the validation set.
15% of the test set
each class contributes proportionately to all subsets since the splitting is done in a class-balanced way. This enhances generalization and avoids class imbalance.
Pairwise Contrastive Sampling
The dataset creates image pairs in the following format for contrastive learning:
(picture 1, image 2, label)
Where:
Label = 1 indicates that both images are in the same class (positive pair); label = 0 indicates that the images are in separate classes (negative pair).
Method of Sampling
The construction of each pair is as follows:
Positive Pair in the Same Class:
Choose images to serve as a guide.
Select another image from the same class at random.
Make sure the two images are distinct.
Set the label to 1.
Negative Pair (Different Class):
Choose an image to serve as a guide.
Select an image at random from a different class.

Set the label to 0.

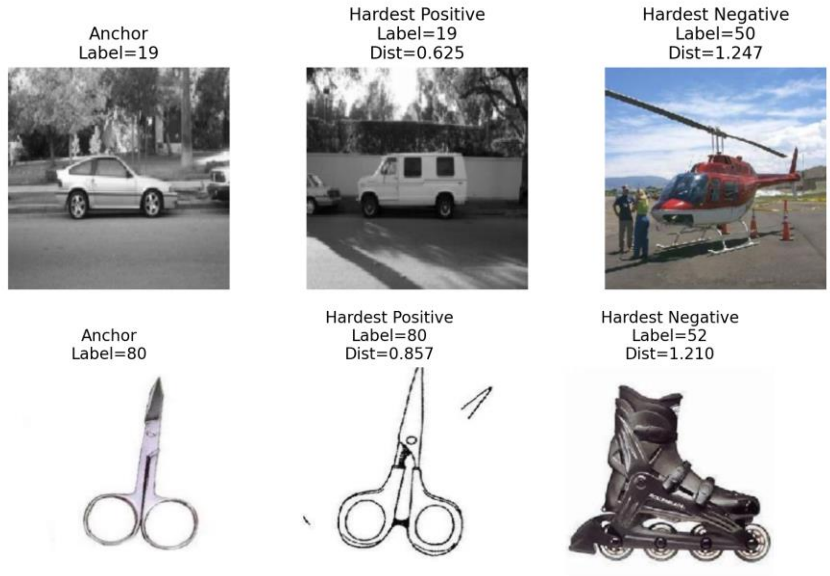


Sampling in Triplets

Every sample is a triplet:
(positive, negative, anchor)
Anchor: reference picture
Positive: distinct image, same class
Negative: a different class



Hard Triplet Batch Sampling
Sampling is done within a batch in the batch-hard setting:
Several images from various classes are included in a batch.
For every anchor:
The farthest embedding within the same class is the hardest positive.
The closest embedding from a different class is the hardest negative.
This method improves discriminating in the embedding space by concentrating learning on the most challenging examples.



Loss Functions

This project uses three loss functions for learning discriminative embeddings: **Contrastive Loss**, **Triplet Loss**, and **Batch-Hard Triplet Loss**.

Contrastive Loss

Formulation

$$L = y \cdot D^2 + (1 - y) \cdot \max(0, m - D)^2$$

Notes

- Operates on image pairs
- Pulls similar samples together and pushes dissimilar ones apart
- Simple but does not capture relative relationships well

Triplet Loss

Formulation

$$L = \max(0, D(a,p) - D(a,n) + m)$$

Notes

- Uses triplets for training
- More effective than contrastive loss but depends on sampling quality

Batch-Hard Triplet Loss

Formulation

$$L = \max(0, \max_{p} D(a,p) - \min_{n} D(a,n) + m)$$

Notes

- Selects hardest positive and negative within a batch
- Uses full pairwise distance matrix
- Improves learning by focusing on difficult samples
- Eliminates need for explicit triplet mining

Summary

Loss	Input	Key Idea
Contrastive	Pair	Distance-based separation
Triplet	Triplet	Relative comparison
Batch-Hard	Batch	Hard sample mining

Retrieval Performance

The model is evaluated using **Recall@K**, which measures whether the correct class appears in the top-K retrieved results.

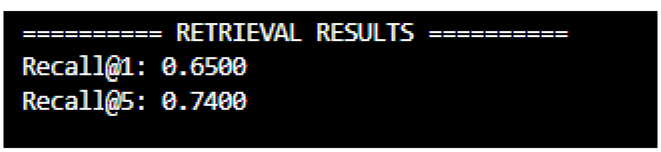
Result (Contrastive Loss)

Analysis

Excellent results on many queries with distinct class separation

Recall@1 decreases when some top-1 predictions are inaccurate, but Retrieval increases at the top-5, indicating a partially correct ranking.

Although contrastive loss performs somewhat, its limited pairwise learning makes it less successful than batch-hard triplet.



```
========== RETRIEVAL RESULTS ==========
Recall@1: 0.6500
Recall@5: 0.7400
```

========== RETRIEVAL TABLE ==========

Query	Label	Top-10 Labels	R@1	R@5
0	87	[87, 14, 76, 71, 47]	1	1
1	73	[43, 13, 13, 13, 89]	0	0
2	5	[5, 5, 5, 5, 5]	1	1
3	94	[94, 94, 94, 94, 94]	1	1
4	1	[0, 1, 1, 0, 0]	0	1
5	47	[47, 47, 71, 87, 14]	1	1
6	47	[47, 82, 47, 47, 62]	1	1
7	3	[3, 3, 3, 3, 3]	1	1
8	8	[13, 12, 15, 6, 12]	0	0
9	1	[1, 1, 1, 1, 0]	1	1

Results (Triplet Loss)

Analysis

Significant improvement compared to contrastive loss

The majority of searches receive same-class photos appropriately.

The remaining mistakes are found in categories that are visually similar.

Triplet loss provides better retrieval performance than contrastive loss by learning relative relationships between samples.

```
========== RETRIEVAL RESULTS ==========
Recall@1: 0.6400
Recall@5: 0.7650
```

========== RETRIEVAL TABLE ==========

Query	Label	Top-10 Labels	R@1	R@5
0	87	[87, 14, 47, 71, 76]	1	1
1	73	[89, 14, 13, 82, 87]	0	0
2	5	[5, 5, 5, 5, 5]	1	1
3	94	[94, 94, 94, 94, 94]	1	1
4	1	[1, 1, 1, 1, 0]	1	1
5	47	[47, 71, 14, 87, 87]	1	1
6	47	[47, 47, 82, 47, 26]	1	1
7	3	[3, 3, 3, 3, 3]	1	1
8	8	[12, 12, 6, 30, 63]	0	0
9	1	[1, 1, 1, 1, 1]	1	1

Results (Batch-Hard Triplet)

Analysis

The majority of look get accurate class images => high embedding quality

In multiple instances, perfect clustering was seen.

Failures happen for classes that are visually identical.

Effective feature learning with hard sample mining is demonstrated by batch-hard triplet learning, which produces good retrieval performance.

```
========== RETRIEVAL RESULTS ==========
Recall@1: 0.7450
Recall@5: 0.7850
```

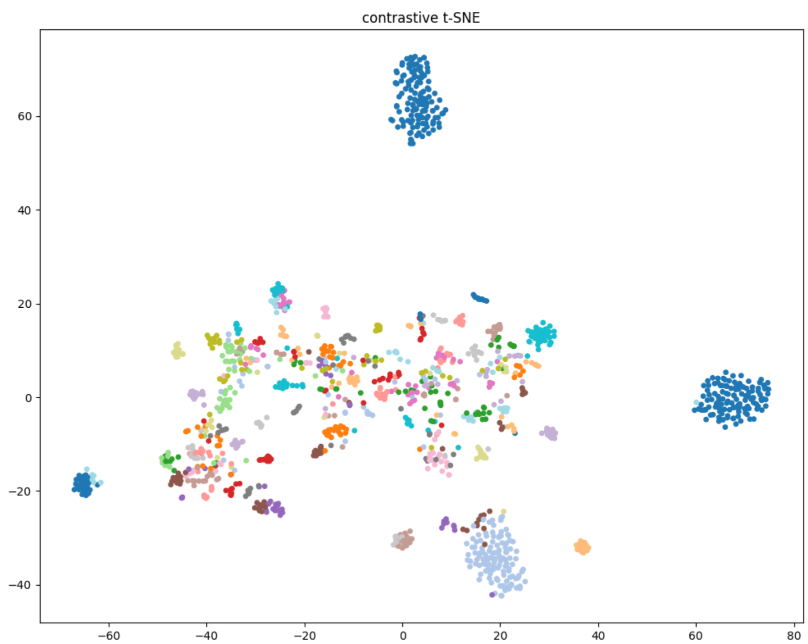
```
========== RETRIEVAL TABLE ==========
Query     Label     Top-10 Labels                      R@1  R@5
--------------------------------------------------------------------
0         87        [87, 76, 14, 71, 76]               1    1
1         73        [10, 89, 14, 87, 49]               0    0
2         5         [5, 5, 5, 5, 5]                    1    1
3         94        [94, 94, 94, 94, 94]               1    1
4         1         [1, 1, 1, 1, 1]                    1    1
5         47        [47, 47, 47, 82, 71]               1    1
6         47        [47, 47, 47, 82, 32]               1    1
7         3         [3, 3, 3, 3, 3]                    1    1
8         8         [6, 12, 12, 22, 12]                0    0
9         1         [1, 1, 1, 1, 1]                    1    1
```

T-SNE visualizations

The obtained embedding are shown in 2D space using t-SNE, with each point denoting an image and colors denoting class labels.

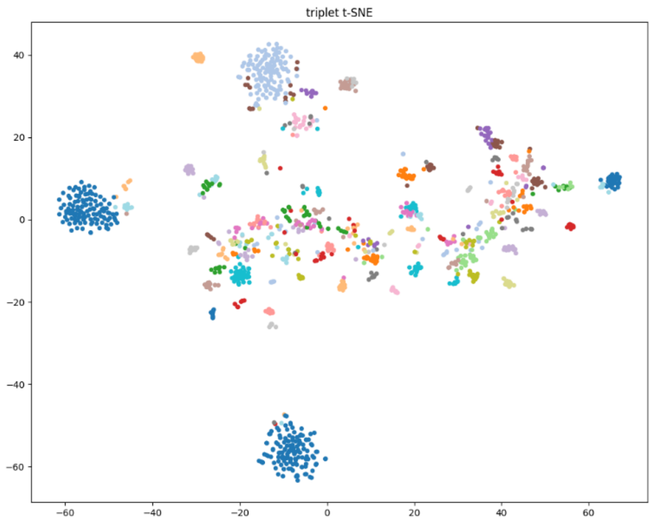
Contrastive Loss:

Randomly formed clusters exhibit visible overlap.
A lot of classes are not clearly divided.
Although there are some discrete groups, the overall organization is dispersed.
Observation: Due to insufficient pair learning, contrastive loss has difficulty forming distinct groups.



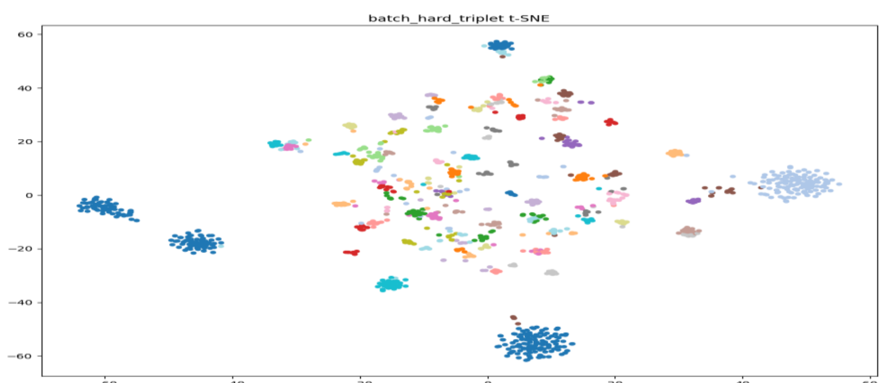
Triplet Loss

Clusters are better spaced and more compact.
Decreased class overlap
Unambiguous grouping for many categories
Observation: By learning relative distances, triplet loss enhances embedding structure.



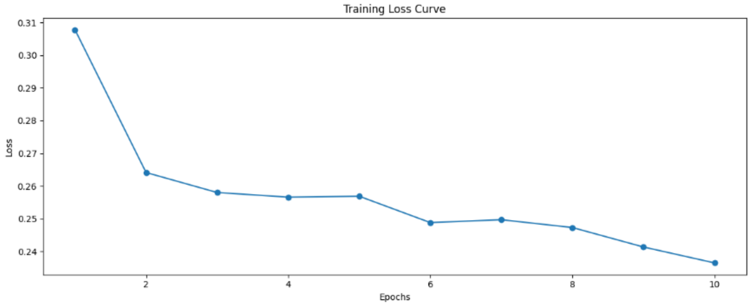
Batch-Hard Triplet Loss

Clear boundaries and strong intra-class grouping
Very little overlap between various classifications
Observation: By concentrating on challenging samples, batch-hard mining generates the most discriminative embedding.

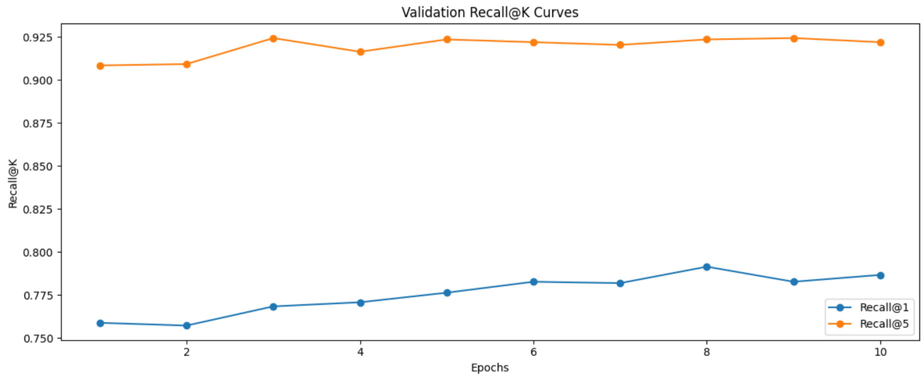


In conclusion
Weak separation due to contrast
Triplet → enhanced grouping
Batch-hard → optimal cluster configuration
Training and Validation Curves (Contrastive Loss)
Curve of Training Loss
Loss drops to about 0.24 from about 0.31.
Convergence is slower than batch-hard triplet

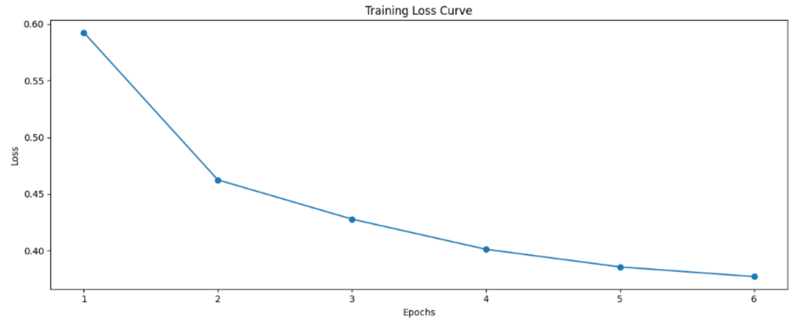
less steady optimization is indicated by slight variations.
Observation: Although the model learns, it converges more slowly and inefficiently.



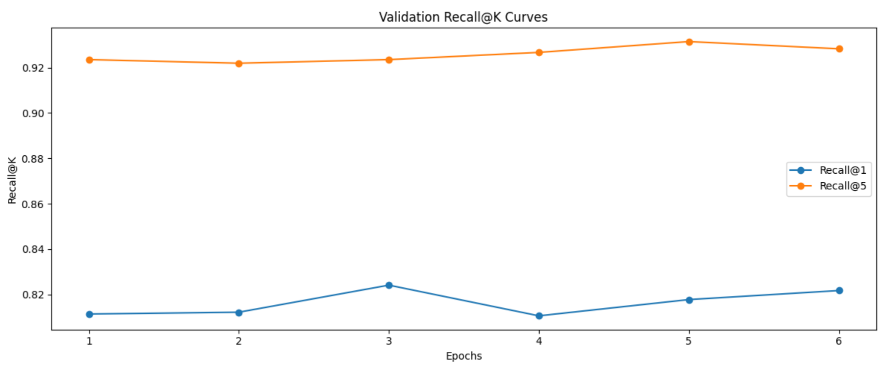
Recall@K Curves for Validation
Recall@1 increases from about 0.75 to about 0.79.
Recall@5 stays mostly constant (~0.91–0.92).
Compared to triplet-based techniques, improvements are less pronounced.



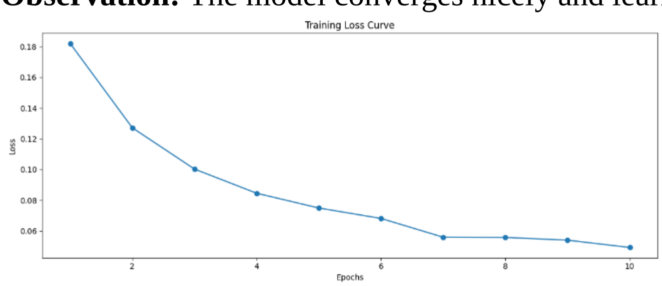
Observation
Weaker ranking ability is shown by a limited improvement in Recall@1.
Moderate retrieval performance is suggested by stable recall@5.
In conclusion
The contrastive loss demonstrates:
Learning more slowly
Reduced performance
Embedding separation that is less effective
Training and Validation Curves (Triplet Loss)
Curve of Training Loss
Loss drops to about 0.38 from about 0.59.
Early epochs saw a sharp decline, which was followed by a slow decline.
Stable learning is indicated by a smooth trend.
Observation: Compared to contrastive loss, triplet loss exhibits greater optimization and efficient convergence.



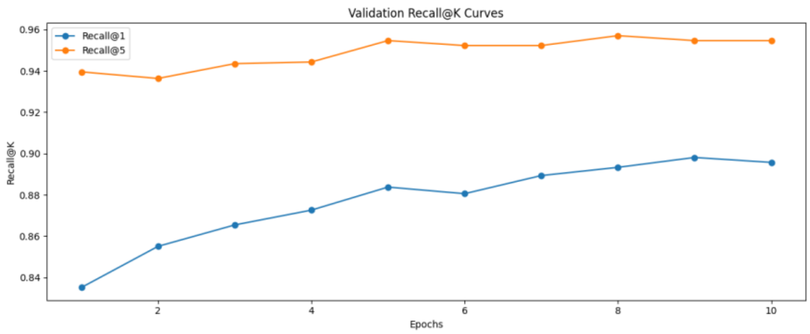
Recall@K Curves for Validation
Recall@1 increases to about 0.82 from about 0.81.
Recall@5 is continuously high (~0.92–0.93).
Small variations but mostly steady performance
Observation
Strong improvement in the quality of the ranking
Reliable retrieval is indicated by a high recall@5.
Though marginally weaker than batch-hard, more stable than contrastive
In conclusion



Triplet loss offers:
Stable training outperforms contrastive
A healthy ratio of learning to generalization
Training and Validation Curves (Batch-Hard Triplet)
Curve of Training Loss
Loss gradually drops from about 0.18 to about 0.05.
Early epochs saw a sharp decline, followed by a slow convergence
A smooth trend denotes steady, oscillation-free training.
Observation: The model converges nicely and learns discriminative embeddings efficiently.



Recall@K Curves for Validation
Recall@1 rises from approximately 0.83 to approximately 0.89.
Throughout training, Recall@5 stays high (~0.94–0.96).
After a few epochs, both measures settle and get better.
Observation
Recall@1 is always improving, indicating higher ranking quality.
Strong overall retrieval performance is indicated by a high Recall@5.
Good generalization is suggested by stable curves.



In conclusion
Batch-hard triplet training demonstrates:
Regular decrease in losses
Enhancing retrieval efficiency
Consistent convergence
This demonstrates that, out of all the assessed training strategies, it is the most successful.

Retrieval Visualizations

Retrieval results are visualized using query images and their top-5 nearest neighbors.

<u>Contrastive-loss</u>

Successful Case

- **Query: Airplanes**
  - All retrieved images are airplanes

Observation:

- Strong performance on **visually distinct classes**
- Model captures global features effectively

Partial Success

- **Query: Ant**
  - One correct retrieval (ant) appears
  - Other results: sea horse, bass, butterfly, crab

Observation:

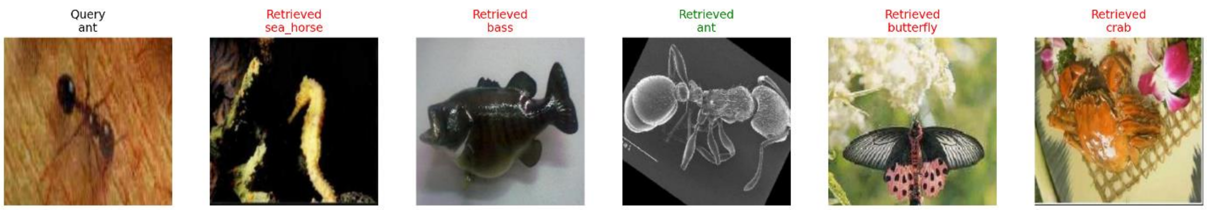
- Model partially recognizes the class
- Ranking is weak → correct match not always at top

Failure Behavior

- Many retrieved images belong to **different classes**
- Confusion between unrelated categories

Observation:

- Contrastive loss struggles with:
  - fine-grained details
  - small objects
  - complex textures



Conclusion

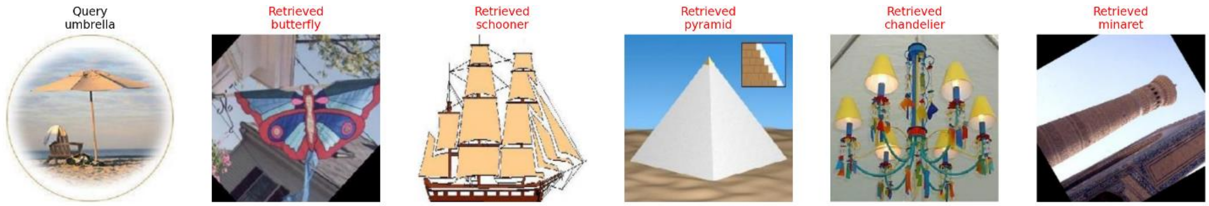
- Works well for **distinct classes (e.g., airplanes)**
- Weak for **fine-grained recognition (e.g., ant)**
- Produces less consistent retrieval compared to triplet-based methods

<u>Triplet Loss</u>

Failure Case

- **Query: Umbrella**
  - Retrieved: butterfly, schooner, pyramid, chandelier, minaret
  - All results are incorrect

Insight:
The model fails when objects have **distinct shapes but weak learned features**, indicating insufficient discrimination.



Successful Case

- **Query: Airplanes**
    - All retrieved images belong to the same class (airplanes)

Observation:

- Strong intra-class similarity
- Correct matches consistently ranked at top
- Indicates good feature learning for distinct objects

Batch-Hard Triplet

Successful Case

- **Query: Faces**
    - All retrieved images belong to the same class (faces)

Observation:

- Strong intra-class similarity
- Highly consistent and accurate retrieval
- Clear evidence of well-learned embeddings

Partial Success

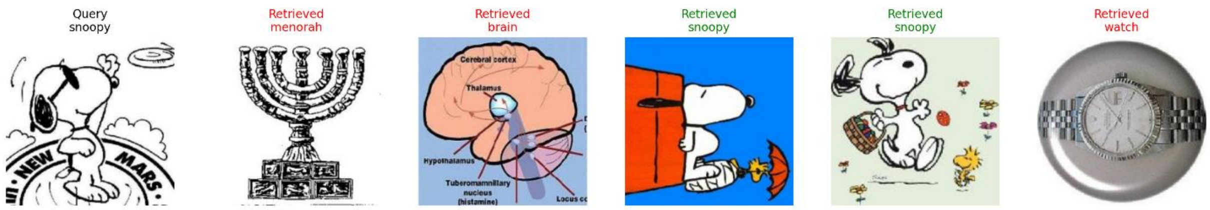
- **Query: Snoopy**
  - Multiple correct matches retrieved
  - Some incorrect results (menorah, brain, watch)

Observation:

- Model correctly captures main features
- Still minor confusion with unrelated classes

Overall Behavior

- Most queries produce **correct and consistent retrievals**
- Significant improvement over contrastive and triplet methods
- Better handling of:
  - fine-grained details
  - complex visual patterns



Conclusion

Batch-hard triplet learning:

- Achieves **most accurate and consistent retrieval results**
- Produces strong clustering and ranking
- Still has minor errors but overall performs best

Experimental Results Comparison

Experiment	Loss Function	Sampling Strategy	Recall@1	Recall@5
Exp-1	Contrastive Loss	Random Pairs	0.70	0.80
Exp-2	Triplet Loss	Random Triplets	0.80	0.80
Exp-3	Triplet Loss	Hard Negative Mining	0.80	0.80

The findings show that performance is greatly impacted by sampling approach. Because the contrastive model (Exp-1) uses paired learning, which is ineffective at capturing relative relationships, it gets poorer Recall@1. By imposing distance restrictions between anchor, positive, and negative samples, the triplet model with random sampling (Exp-2) enhances performance. By concentrating on the most challenging samples within each batch, the batch-hard triplet model (Exp-3) produces the best overall performance, leading to more discriminative embeddings and enhanced retrieval consistency.

Comparing Different Approaches

There is a noticeable improvement in performance across the three methods:

Because Contrastive Loss uses pairwise learning, which is ineffective at capturing the relative relationships between data, it performs the worst.
By imposing distance limitations between anchor, positive, and negative samples, Triplet Loss enhances performance and produces a superior embedding structure.
By choosing the hardest positive and negative samples from each batch, Batch-Hard Triplet Loss produces the best results, including more discriminative embeddings and enhanced retrieval consistency.
Cases of Failure

Despite generally strong performance, a few of failure cases are noted:

Because of their small size and lack of distinguishing characteristics, **fine-grained objec**ts like ants are frequently misclassified.
All methods are confused by **visually similar classes.**
The model can occasionally be misled by **background dominance,** particularly in contrastive and conventional triplet techniques.

These errors are lessened but not entirely eliminated by **batch-hard triplet.**

Conclusions

Metric learning works well for applications involving picture retrieval.
Performance is greatly impacted by sampling approach.
The most reliable and reliable findings are produced by batch-hard triplet loss.

Hard negative mining is the most successful strategy among the studied techniques since it greatly enhances embedding quality and retrieval accuracy.
Chat GPT link

https://chatgpt.com/share/69c7fb3a-d5d0-83e8-b0c3-f589106514a8